

Soft Computing-Based EEG Classification by Optimal Feature Selection and Neural Networks

Muhammad Hamza Bhatti , Javeria Khan , Muhammad Usman Ghani Khan, Razi Iqbal ,
Moayad Aloqaily , Yaser Jararweh, and Brij Gupta 

Abstract—Brain computer interface translates electroencephalogram (EEG) signals into control commands so that paralyzed people can control assistive devices. This human thought translation is a very challenging process as EEG signals contain noise. For noise removal, a bandpass filter or a filter bank is used. However, these techniques also remove useful information from the signal. Furthermore, after feature extraction, there are such features which do not play any significant role in effective classification. Thus, soft computing-based EEG classification followed by extraction and then selection of optimal features can produce better results. In this paper, subband common spatial patterns using sequential backward floating selection is being proposed in order to classify motor-imagery-based EEG signals. The signal is decomposed into subband using a filter bank having overlapped frequency cutoffs. Linear discriminant analysis followed by common spatial pattern is applied to the output of each filter for features extraction. Then, sequential backward floating selection is applied for selection of optimal features to train radial basis function neural networks. Two different datasets have been used for evaluation of results, i.e., Open BCI dataset and EEG signals acquired by Emotiv Epoc. The proposed system shows an overall accuracy of 93.05% and 85.00% for both datasets, respectively. The results show that the proposed optimal feature selection and neural network-based classification approach with overlapped frequency bands is an effective method for EEG classification as compared to previous techniques.

Index Terms—Electroencephalogram (EEG), linear discriminant analysis (LDA), radial basis function neural networks (RBFNN), sequential backward floating selection (SBFS), soft computing.

Manuscript received March 4, 2019; revised May 7, 2019; accepted May 29, 2019. Date of publication June 28, 2019; date of current version October 3, 2019. Paper no. TII-19-0674. (Corresponding author: Moayad Aloqaily.)

M. H. Bhatti and J. Khan are with the Al-khawarizmi Institute of Computer Science, University of Engineering and Technology, Lahore 54000, Pakistan (e-mail: muhammad.hamza@kics.edu.pk; javeria.khan@kics.edu.pk).

M. U. G. Khan is with the Department of Computer Science, University of Engineering and Technology, Lahore 54890, Pakistan (e-mail: usman.ghani@kics.edu.pk).

R. Iqbal is with the American University in the Emirates (AUE), Dubai 503000, United Arab Emirates (e-mail: razi.iqbal@ieee.org).

M. Aloqaily is with the Gnowit Inc., Ottawa, ON K2K 1Y6, Canada (e-mail: moayad@gnowit.com).

Y. Jararweh is with the Jordan University of Science and Technology, Irbid 3030, Jordan (e-mail: yjararweh@just.edu.jo).

B. Gupta is with the National Institute of Technology Kurukshetra, Kurukshetra 136119, India (e-mail: bbgupta@nitkkr.ac.in).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TII.2019.2925624

I. INTRODUCTION

THE HUMAN brain is based on billions of neurons [1]. Information about a specific command is transferred from the human brain to other body parts when these neurons communicate with each other. This communication of neurons produces electrical signals as a result of various brain activities, i.e., thought process, emotions or any other brain activity [2]. These signals are known as electroencephalogram (EEG) signals. EEG signals can be acquired by using two different ways, i.e., invasive and noninvasive acquisition. For data acquisition by invasive technique, EEG electrodes should be placed inside the human brain by extensive surgeries. While the noninvasive technique of data acquisition uses wet or dry electrodes that are placed on the scalp for the direct acquisition of EEG signals. Although the invasive technique has good signal strength, it can be harmful due to post-surgery infections, whereas the noninvasive technique is advantageous in terms of safety but the signal strength is weak as compared to invasive techniques [3]. Therefore, due to ease and inexpensiveness, the noninvasive technique has been widely used for brain computer interface (BCI) for controlling various BCI applications, i.e., neuro-gaming, prosthetics, wheelchair, etc. BCI translates electrical signals into control commands which can be used for patients suffering from paralysis, as their brain remains unaffected [4]. For paralyzed people, signals produced as a result of imagination have proven to be very effective. Motor-imagery (MI) signals are produced as a result of the imagination of motor movements, which can be acquired from the motor cortex of the human brain. This system of human thought translation is done by implementing various soft computing techniques, after cleaning and feature extraction of the acquired signal. Soft computing consists of those computational machine learning techniques where approximated, imprecise, and uncertain results are tolerated for those problems which do not have an exact solution. This computing technique is widely used for EEG classification because the acquired signals have low signal strength, which provides a provision of some error in the classification results of a BCI system.

As acquired EEG signal consists of artifacts produced as a result of certain involuntary actions, i.e., eye blinking artifact, eye movements, signals due to muscle movement, etc. Therefore preprocessing is done to remove these artifacts using various preprocessing techniques, i.e., bandpass filter at a specific frequency band or averaging of the signal by using spatial filters [5]. After preprocessing of EEG signals, different features are extracted for effective classification. In literature, different

feature extraction techniques have been used to extract desired features from the acquired EEG data. Common-spatial pattern (CSP) [6] method is a feature extraction technique, which has proved its efficiency in topographic pattern extraction of brain data [7]. In general, CSP can be applied on noisy or cleaned EEG data but the selection of poor frequency bands will result in poor recognition accuracy of the system. Usually, a single wide band having a range of 4 to 40 Hz is used to filter the desired band from the MI-based EEG signal. Due to this wide band filter, different undesired noisy components are also added to the filtered EEG signal on which the CSP is to be applied.

Selective search and manual removal for each subject is an effective way to find the best frequency band. These processes are time consuming and meticulous but produce effective results. Less standardization in selective selection is also a major concern. To overcome this problem, different variants of CSP have been proposed, i.e., common spatio-spectral pattern (CSSP) [8], common sparse-spectral-spatial pattern (CSSSP) [9], and subband common-spatial patterns (SBCSP) [10]. In CSSP, spatial filters along with simple delayed filters are optimized together. While CSSSP proposed the CSP analysis along with arbitrary finite impulse response filter [9] optimization. CSSSP produces better results than CSSP because it computes the spectral patterns for all channels instead of computing for each channel separately. For better accuracy in case of MI brain signals, another variant of CSP named as SBCSP has been proposed. In SBCSP, decomposition of EEG signal into different subband is done by passing it from filter bank which consists of bandpass filters for different frequency cutoffs. Then, linear discriminant analysis (LDA) followed by CSP is applied on the decomposed signal and the resulted transformed features of LDA are then fused to train the classifier on these features. Although SBCSP has shown better results than CSP, CSSP, and CSSSP, this technique ignored the possible importance of the features produced by subband. To overcome this problem of SBCSP another variant of CSP, filter bank common-spatial pattern (FBCSP) was proposed [11]. In FBCSP, for selecting the optimal features, the mutual information of extracted features followed by different subband filter bank is computed. Then the classifier is trained on the selected features. FBCSP achieved better results than SBCSP but it also uses the filter bank having different subband of user-defined frequency gaps. Wavelet packet transform (WPT) [12] is also used for EEG signal decomposition rather than filter bank. In this technique, the signal is decomposed into approximated and detailed coefficients until the desired frequency domain components are not obtained. Then these components are provided to the radial basis function neural networks (RBFNN) [13] classifier as an input feature vector. EEG decomposition by WPT produces better results than FBCSP but the major drawback in this technique is that it selects only specific frequency components from the whole signal which causes loss of important information from EEG signal.

Another variant of CSP, discriminant filter bank common-spatial pattern (DFBCSP) [14] was introduced to overcome the deficiency mentioned in FBCSP and WPT. DFBCSP selects the

most discriminant subband of filter bank for single channel, i.e., C3, C4, Cz, etc. on the basis of the Fisher ratio [14]. After selection of subband, CSP is applied on each subband for feature extraction and then the training of the classifier is done on the extracted features. DFBCSP showed less computational cost than FBCSP and improved results than both of the FBCSP and SBCSP. For further improved results, sparse filter bank common-spatial pattern (SFBCSP) was proposed [15]. SFBCSP uses multiple filter bands to optimize sparse filter patterns and CSP features selection from multiple bands is done by using a newly defined “Sparse Filter Band” supervised technique [15]. At the end, support vector machine (SVM) [16] is used for classification of the results on the basis of extracted features. An improved discriminant filter-bank selection approach for MI-based brain signal classification [17] was also proposed for the selection of optimal bands in filter bank on the basis of mutual information computed for each subband. This approach is applied to the data of all channels instead of one channel as described in DFBCSP. This approach also includes a wide band filter of 6–30 Hz along with other features for 12 overlapped subband, i.e., in a range of 4–30 Hz having a band gap of 4 Hz. SVM was used to calculate the misclassification rate. In this paper, the overall accuracy was improved from DFBCSP but due to double computation, the computation cost of the system is also greater than the other variants of CSP, i.e., first of all, the selection of optimal filter bands and the whole process of feature-extraction then selection and classification. As EEG data of all the channels is used in this approach, therefore, there is a high possibility of less accuracy rate due to the artifacts included in EEG data, i.e., eye blinking artifacts which are captured by the electrodes placed on the frontal and occipital lobe of the brain. In current researches, sparse Bayesian learning is also being used for optimal feature selection [18]. In this approach, the decomposition and feature extraction are done in a similar way as in previous approaches, i.e., by using filter bank having different subband and CSP, respectively but SVM is trained on the sparse features obtained by using Bayesian learning technique. **Table I** shows a brief comparison of previous work which involves different soft computing-based classification of MI brain signals for the imagination of right and left-hand movements.

In SBCSP and FBCSP, the filter bank consists of six subband filters having the cutoffs in a range of 8 to 32 Hz with a user defined gap of 4 Hz [10], [11]. As this frequency gap in bands is user defined, which can be any number so, this can also be the reason behind the loss of some useful information (information, which can be in between two selected bands) from the EEG signal [17]. Therefore, the information which lies in between two subband will be divided and may not be useful at all. While in DFBCSP and improved DFBCSP, the used filter bank consists of overlapped frequency bands between 6 and 30 Hz along with a wide frequency band of 6–30 Hz [14], [17] have not shown effective results. So, in the proposed approach, instead of using a filter bank which has six different filter bands or 12 overlapped frequency bands with a wide range frequency band, the filter bank which consists of 12 filters of two different categories having different frequency cutoffs between 6 and

TABLE I
COMPARISON OF RELATED WORK IN THE LITERATURE

Technique	No. of Channels	Dataset	Feature Extraction	Feature Selection	Classifier	Accuracy
Lotte et al. 2011 [6] (CSP)	118	Dataset IVa BCI competition III	CSP	-	SVM	80.43 %
Lemm et al. 2005 [8] (CSSP)	118	Dataset IVa BCI competition III	CSP	-	SVM	81.67 %
Dornhege et al. 2005 [9] (CSSSP)	118	Dataset IVa BCI competition III	CSP	-	SVM	80.51 %
Novi et al. 2007 [10] (SBCSP)	118	Dataset IVa BCI competition III	Sub-band CSP & LDA	RFE	SVM	87.40 $\pm$ 1.30 %
				MC		84.40 $\pm$ 1.30 %
Kai Keng Ang et al. 2008 [12] (FBCSP)	118	Dataset IVa BCI competition III	Sub-band CSP	MINBPW	NBPW	86.30 $\pm$ 1.10 %
Khare et al. 2011 [13] (WPT)	8	Self-Acquisition	WPT	-	RBFNN	87.96 $\pm$ 0.60 %
Higashi et al. 2013 [15] (DFBCSP)	118	Dataset IVa BCI competition III	CSP on Single EEG Channel	-	LDA	89.06 $\pm$ 2.43 %
Zhang et al. 2015 [16] (SFBCSP)	118	Dataset IVa BCI competition III	CSP & Sparse Filter Bands	-	SVM	91.05 $\pm$ 2.45 %
Kumar et al. 2017 [18] (Improved DFBCSP)	118	Dataset IVa BCI competition III	CSP & LDA	-	SVM	91.68 $\pm$ 4.48 %
Zhang et al. 2017 [19] (SBL)	64	Self-Acquisition	CSP	SBL	SVM	90.87 $\pm$ 5.23 %
Proposed Algorithm (SBCSP-SBFS)	14 & 118	Emotiv Epoc & Dataset IVa BCI competition III	Sub-band CSP	SBFS	RBFNN	85.00 $\pm$ 0.8 % (Emotiv Epoc) 93.05 $\pm$ 1.4 % (BCI Dataset)

32 Hz is used. Each category in filter bank has six filters, cutoffs of the filters in each category are defined in such a way that the frequencies of both two categories should overlap [14]. The first category of filter bank has a gap of 4 Hz between 8 to 32 Hz while the second category has also a gap of 4 Hz but it is in between 6 and 30 Hz. As the gap between two frequency bands is 4 Hz and there is also a difference of 2 Hz between the starting frequencies of both categories, therefore, there is an overlapping of 2 Hz between each band in both categories. The major benefit of this overlapping is the preservation of information which can be divided into two different frequency bands due to sharp frequency cutoffs. As this overlapping occurs between two consecutive bands so, half portion of each band is overlapped and the whole divided information is preserved. If in case some information is divided due to sharp frequency cutoffs, then this preserved information could be used as it is required.

Another shortcoming in previously used techniques is the method for feature selection. In SBCSP, the extracted features are fused before passing them to the classifier for training. In other words, this score fusion is the selection of those optimal features from the extracted features on which the classifier gives the best accuracy. As EEG signals contain noise in them which is also present in the computed subband features therefore only a few bands will be useful and the selection of these bands or corresponding computed features to these bands

will produce better results. For this purpose of score fusion, recursive feature elimination (RFE) technique or meta classifier (MC) were used [10]. Although these techniques produce good results, but in every run, these methods select different features on the basis of accuracy, which make them unsuitable for effective feature selection. Due to this random feature selection, these methods also produce unstable classification results. So, sequential backward floating selection (SBFS) [19] is used in the proposed approach to select the optimal features from features of all subband, as well as to overcome the problem of instability in results. Therefore, this paper proposes an improved SBCSP variant of CSP. The improvements which make the proposed approach unique in previously described approaches are the addition of a wide range of overlapped frequency bands in filter bank and the selection of optimal features by using SBFS.

The rest of this paper is organized as follows. Section II describes methodology while Section III provides a detailed discussion on results. Section IV concludes this paper.

II. METHODOLOGY

In the proposed algorithm, first, a filter bank is used to decompose the EEG signal into subbands. This filter bank consists of different bandpass filters having different frequency bands. In this case, the filter bank consists of 12 filters of two

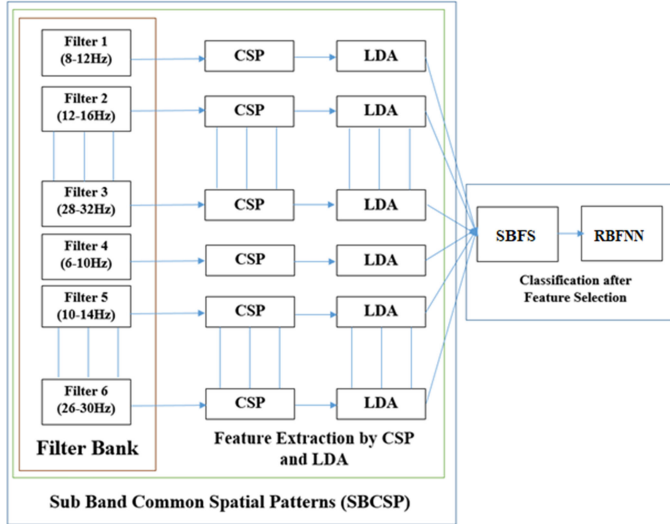


Fig. 1. Block diagram of the proposed approach.

different categories having different frequency cutoffs between 6 and 32 Hz is used. Each category in filter bank, cutoffs of the filters in each category are defined in such a way that the frequencies of both two categories should overlap. The first category of filter bank has a gap of 4 Hz between 8–32 Hz while the second category has also a gap of 4 Hz but it is in between 6–30 Hz. Subsequently, the CSP technique is applied to the output of each bandpass filter to extract the features. These features are then fed into LDA [20] to obtain scores which signify the classification accuracy of each frequency band. Then, SBFS is applied to select the optimal features from features of all subband resulted by LDA. This selection of features is done on the basis of the overall accuracy of the system for different combinations from extracted subband features. Finally, RBFNN [13] is used for classification on the basis of selected features. Figs. 1 and 2 represent the block diagram and algorithm of the proposed approach respectively.

A. Common Spatial Pattern

CSP is used for extracting the features which are used in the classification of two class data in a dataset. It also produces satisfactory results by extracting significant EEG features for MI-based brain signal classification. Learning the linear transforms (or spatial filters) for maximizing the ratio of transformed data variance between different classes in EEG data, is the objective of CSP.

Let us assume two classes of EEG data which have been centered without loss of generality, i.e., $Y(i, 1)$ and $Y(i, 2)$ captured and stored in the form of (channels * number of samples), where i = specific trail in EEG data. The spatial covariance matrix Σ for “ m ” classes is computed by using (1). While, $m = 1, 2$

$$\Sigma_m = \frac{1}{N_m} \sum_{i=0}^{N_m} Y_{i,m} Y_{i,m}^T \quad (1)$$

1. **Input:** Raw EEG data/ EEG Epoch
2. **Output:** Predicted Class of EEG Signal

Algorithm 1 Sub-band Common Spatial Patterns using Sequential Backward Floating Selection

```

1: function SBCSP-SBFS ( )
2:   eeg_data = (N * C)    ▷ Where N - Number of Samples in EEG Data
                           ▷ C = Number of Channels
3:
4:   win_size = t = 2sec
5:   f_s = 128Hz           ▷ Where f_s - Sampling Frequency
6:   epoch_size = win_size * f_s
7:   eeg_data, label = Separate_Data (raw_data[i, :])
8:   eeg_data_epoch = Create_Epoch (eeg_data, epoch_size)
9:   for i = 1 to [length (eeg_data_epoch)] do ]
10:    sbfe = FilterBank (eeg_data_epoch, freq_cut)
11:    sbfp = CSP (sbfe)
12:    lda_scores = LDA (sbfp)
13:  end for
14:  for n combinations of lda_scores do
15:    selected_features = SBFS (lda_scores)
16:  end for
17:  train_data, train_label, test_data, test_label = Split_Data
    (selected_features, split_ratio)
18:  classifier_model = RBFNN (train_data, train_label)
19:  predicted_class = classifier_model (test_data)
20:  return predicted_class
21: end function

```

Fig. 2. Proposed algorithm.

where

N_m = Total number of trials in “ m ” classes

According to the major principle of CSP, as described above, the transformed data variance between two classes is calculated by using (2)

$$\max_v J(v) = \frac{v^T \Sigma_1 v}{v^T \Sigma_2 v} \quad \text{s.t. } \|v\|_2 = 1 \quad (2)$$

where

$J(v)$ = Maximization of the Rayleigh quotient

v = Spatial filter

$\|v\|_2$ = The m_2 normal

J_v is calculated by using (3), i.e., solving the generalized eigenvalue equation

$$\Sigma_1 v = \lambda \Sigma_2 v. \quad (3)$$

The eigen vectors are then computed from the learned spatial filter matrix V . The projection of an EEG signal can be computed as follows:

$$U = V^T X \quad (4)$$

where

X = EEG sample

V = Learned spatial matrix

U = CSP projections matrix of EEG sample, computed from learned spatial matrix.

B. Linear Discriminant Analysis

LDA is also used to find the projection matrix and this computed matrix can further be used as a feature to train the classifier [20]. LDA finds the cost function for projection matrix

as follows:

$$O = \frac{V_{lda}^T S_B V_{lda}}{V_{lda}^T S_W V_{lda}} \quad (5)$$

where

O = EEG signal's cost function for LDA matrix

S_B = Variance between MI left and right-hand classes

S_W = Variance within MI Left or right-hand classes

V_{lda} = Projection matrix which guarantees maximum separability by maximizing the ratio of S_B to S_W

S_B and S_W are computed by using (6) and (7), respectively

$$S_B = (e_2 - e_1)(e_2 - e_1)^T \quad (6)$$

$$S_W = \sum_{g_p \in m_1} (g_p - e_1)^2 + \sum_{g_p \in m_2} (g_p - e_2)^2 \quad (7)$$

where

e_1 and e_2 = Empirical class means of CSP computed features

m_1 and m_2 = Class 1 and Class 2, respectively.

In the end, the one-dimensional score produced by LDA can be computed as follows:

$$P = V_{lda}^T g_p. \quad (8)$$

C. Sequential Backward Floating Selection

After computation of LDA scores followed by CSP for each subband, these scores are used as a feature for training the classifier. As EEG signals contain noise in them which is also present in the subband features therefore, only a few bands will be useful and the selection of these bands or corresponding computed features to these bands will produce better results. Different techniques are used for optimal feature selection. While in the proposed approach, SBFS is used for the purpose of optimal feature selection, these features will further be used for training of RBFNN classifier. In SBFS, “ k ” optimal features are selected among “ n ” number of total features in such a way that, the classifier should give maximum accuracy on these selected “ k ” features. Let us assume “ B ” is a set of all features, i.e.,

$$B = \{b_1, b_2, \dots, b_n\}.$$

The SBFS technique takes “ B ” set as an input and the selection of output set “ C ” (i.e., subset of set “ B ”) from the set “ B ” is done in a way that first, the classifier is trained on all the features, then one by one those features are eliminated from the set which reduces the overall accuracy of the system. This process of feature elimination is done until the desired number of features are not remained in the feature vector. In the end, this method returns a set “ C ” of optimal features having a length equal to “ k ” desired or to be selected number of features, i.e.,

$$C = \{c_j \mid j = 1, 2, \dots, k; c_j \in B\}$$

where

$$k = (0, 1, 2, \dots, n - 1).$$

D. Radial Basis Function Neural Networks

RBFNN is a 2-layer network which has an input layer fully connected to a hidden layer. Then, the output of the hidden layer

performs a weighted sum of input features to get the output. RBFNN learns to approximate the underlying trend using many Gaussian curves [13]. The major difference between RBF net and regular neural network is that it uses Gaussian RBF inside the hidden layer neurons. RBF nets are used for regression or function approximation. The mathematical form of RBFNN working can be learned.

Given an input vector x , RBF neural network produces a weighted sum output as follows:

$$F(x) = \sum_{j=1}^k w_j \varphi_j(x, c_j) + b \quad (9)$$

where

w_j = weights

b = Bias

k = The number of centers

φ_j = Gaussian RBF

φ_j can be calculated as follows:

$$\varphi_j(x, c_j) = e^{\left(\frac{-\|x - c_j\|^2}{2\sigma_j^2}\right)} \quad (10)$$

where

σ = Standard deviation.

Basic steps for implementation of RBFNN are as follows.

- 1) Scale the extracted features.
- 2) Input vector mapping to a higher dimension.
- 3) Compute the final hidden layer weights.
- 4) Test the network.

III. RESULTS AND DISCUSSION

Two different datasets are used for evaluation of proposed algorithm, i.e., Open BCI Dataset IVa Competition III [21] and Raw EEG signals acquired by Emotiv Epoc. The accuracy results of the proposed system are evaluated by changing the number of selected features. EEG data of those channels is excluded which contain eye artifact, i.e., electrodes placed at frontal lobe.

A. Dataset Description

Open BCI Dataset IVa of Competition 3 is used for the result comparison of the proposed algorithm with previous techniques. The described dataset was recorded from five healthy subjects by using 118 channels headset. This dataset includes MI signals produced as a result of imagining right hand and foot movement. The acquired EEG data from each subject consisted of 280 trials that were sampled at 100 Hz.

Data acquired by using Emotiv Epoc contains EEG signals produced as a result of the imagination of two different tasks, i.e., right and left-hand movements. To acquire the data, 22 years old healthy male subject was shown a “specific stimulus” for 2 s followed by a 2 s “+ sign” and 4 s “blank screen” indicating rest state. The stimulus used for a specific task is represented as an arrow showing the direction of the task, i.e., left or right to the user. A total number of trails for dataset acquisition in the acquired dataset were 160 while, the sampling frequency of

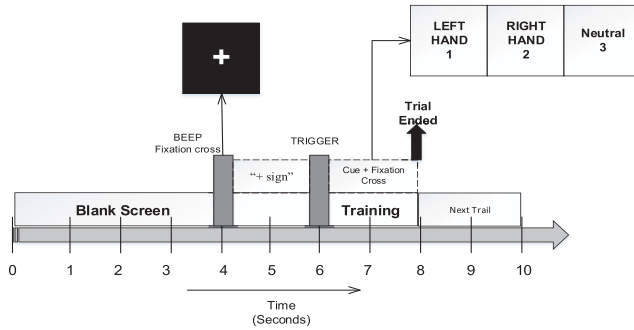


Fig. 3. Data acquisition scheme.

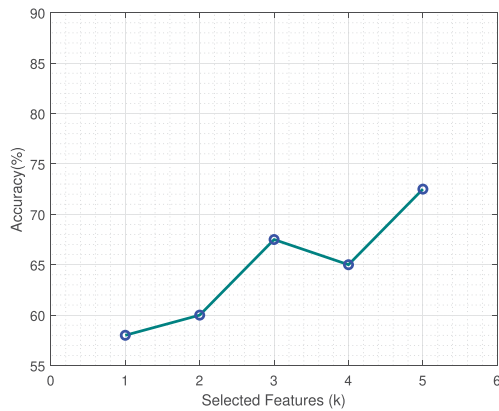


Fig. 4. Graphical comparison of accuracies for subband filter bank by RBFNN-SBFS.

data was 128 Hz. Fig. 3 shows the complete data acquisition scheme, used to acquire EEG data by using Emotiv Epoc.

First of all, six subband features are calculated from the acquired signal. As the EEG signals contain noise in them which is also present in the computed subband features therefore, the selection of optimal bands is done by evaluating the accuracy of the system for different combinations of extracted features (i.e., five combinations for six features). Further, 12 overlapped frequency bands instead of six bands in filter bank produces more number of subband features which provides more optimal features. Figs. 4 and 5 represent the graphical comparison of results produced by the proposed system with a different number of frequency bands in the filter bank. These results are shown for the EEG data acquired by using the Emotiv Epoc headset. Fig. 4 shows the resulted accuracy for six distinct frequency bands in such a way that, subband CSP with RBFNN-SBFS shows a maximum of 72.5% accuracy on the EEG signal produced as a result of the imagination of left and right-hand movement. These results are shown by the proposed system for five selected features for the SBFS method with RBFNN as a classifier. The results show that there is an increase of 5% in the overall accuracy of the proposed system as compared to the overall accuracy for previously implemented techniques. The addition of six new overlapped frequency bands is the main difference between the two approaches experimented in Figs. 4 and 5. Fig. 5 shows the classification accuracy by the proposed approach for 12 overlapped frequency bands in the filter bank

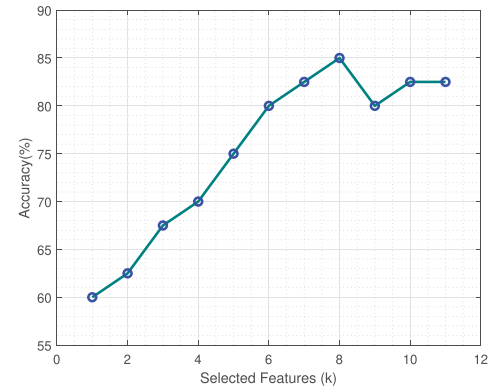


Fig. 5. Graphical comparison of accuracies for overlapped subband filter bank by RBFNN-SBFS.

TABLE II
COMPARISON OF RESULTS AMONG SUBBAND AND OVERLAPPED SUBBAND FOR SBCSP-SBFS

Selected Number of Features (k)	Results for Sub-band (%)	Results for Overlapped Sub-band (%)
1	58.0	60.0
2	60.0	62.5
3	67.5	67.5
4	65.0	70.0
5	72.5	75.0
6	-	80.0
7	-	82.5
8	-	85.0
9	-	80.0
10	-	82.5
11	-	82.5

in a way that, subband CSP with RBFNN-SBFS followed by the new filter bank shows a maximum of 85.0% accuracy on the EEG signal which is produced as a result of the imagination of movement. This accuracy rate is shown by the proposed system for eight selected features for the SBFS method with RBFNN as a classifier. Table II provides the accuracy comparison among subband and overlapped subband for SBCSP-SBFS.

The misclassification error rate is compared for previously implemented techniques in the literature, as shown in Table III. It provides the comparison for both of the datasets consisting of EEG data for multiple subjects, i.e., “aa,” “al,” “av,” “aw,” “ay,” and “sub.” DFBCSP with subject “al,” improved DFBCSP with the subject “aa” and “aw,” and proposed approach with the subject “av” and “ay” produces more accurate results than other techniques. As other techniques also perform well for BCI dataset so, it is hard to say that the proposed approach outperforms. While the average misclassification error by the proposed approach shows that, the overall accuracy is improved by 1.67% for open BCI dataset. Furthermore, dataset acquired by using Emotiv Epoc is tested for all the techniques including the proposed approach; which shows that DFBCSP and FBCSP produce similar results to each other and the improved DFBCSP

TABLE III
MISCLASSIFICATION RATE (%) OF DIFFERENT TECHNIQUES FOR OPEN BCI DATASET AND EMOTIV EPOC DATASET

Subject	CSP(%)	CSSSP(%)	SBCSP(%)	FBCSP(%)	DFBCSP(%)	Improved DFBCSP(%)	Proposed Approach(%)
Dataset IVa from BCI Competition III Techniques							
aa	21.00	15.65	10.70	17.14	9.64	8.79	11.11
al	3.86	2.84	1.40	1.29	1.00	1.14	5.56
av	28.29	28.59	29.60	30.36	31.21	24.05	8.33
aw	10.36	5.72	4.30	6.50	4.64	3.21	5.56
ay	3.86	6.21	4.32	5.07	8.21	4.43	4.17
Average	13.47	13.49	10.00	12.07	10.94	8.32	6.95
Dataset Acquired from Emotiv Epoc							
Sub	37.74	33.33	31.99	28.30	28.57	22.22	15.00

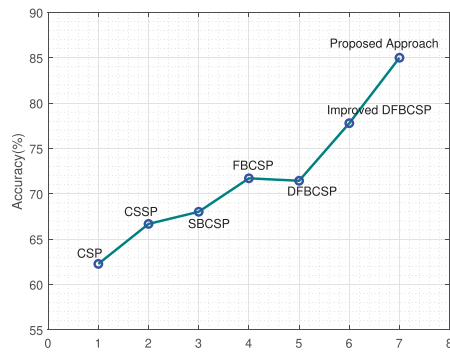


Fig. 6. Graphical comparison of accuracies for subband filter bank by RBFNN-SBFS for Emotiv dataset.



Fig. 7. Graphical comparison of accuracies for subband filter bank by RBFNN-SBFS for BCI competition dataset.

produces 22.22% misclassification error which is minimum among all the previous techniques. While the proposed approach results 15.00% misclassification error which indicates that the accuracy of the proposed algorithm is increased by 7.22% as compared to other techniques. The decrease in misclassification rate for Emotiv Epoc shows a remarkable improvement in accuracy of the proposed algorithm, i.e., 85%. But there is still a need to improve the accuracy because the MI signals acquired from Emotiv Epoc has not good strength as compared to EEG signal in BCI datasets. The environment during data acquisition is also not as sophisticated as for BCI competition datasets. The major reason behind which is that Epoc has only two electrodes on the motor cortex. However a greater number

of electrodes should be placed on the motor cortex to capture the signal produced as a result of imagination. Figs. 6 and 7 show the graphical results comparison of the proposed approach with other techniques for BCI Competition and Emotiv Epoc dataset, respectively.

IV. CONCLUSION

In this paper, an improved SBCSP variant of CSP was introduced in order to improve soft computing-based classification results of MI EEG signals. The improved results showed that SBFS with RBFNN classifier was an effective feature selection method than RFE feature selection technique with SVM classifier used for SBCSP. Further, the addition of overlapped filter banks in the proposed system had shown significant improvements in the results. Result comparison of different previously proposed techniques for two different datasets showed that the proposed approach outperforms for Emotiv Epoc dataset. So, it can be concluded that the proposed approach produced 93.05% accuracy for open BCI dataset and 85.00% accuracy for the dataset acquired in the regular nonsophisticated environment, i.e., real-time data acquisition by Emotiv Epoc. These results showed an increase of 1.67% and 7.22% in the overall accuracy for open BCI and Emotiv Epoc datasets, respectively, as compared to previously implemented techniques in the literature. We are currently working on the improvement of results by introducing efficient algorithms for optimal selection of EEG channels.

REFERENCES

- [1] S. Herculano-Houzel, "The human brain in numbers: A linearly scaled-up primate brain," *Frontiers Human Neurosci.*, vol. 3, 2009, Art. no. 31.
- [2] M. A. Jatoti, N. Kamel, A. S. Malik, I. Faye, and T. Begum, "A survey of methods used for source localization using EEG signals," *Biomed. Signal Process. Control*, vol. 11, pp. 42–52, 2014.
- [3] T. Ball, M. Kern, I. Mutschler, A. Aertsen, and A. Schulze-Bonhage, "Signal quality of simultaneously recorded invasive and non-invasive EEG," *Neuroimage*, vol. 46, no. 3, pp. 708–716, 2009.
- [4] J. R. Wolpaw, N. Birbaumer, D. J. McFarland, G. Pfurtscheller, and T. M. Vaughan, "Brain–computer interfaces for communication and control," *Clin. Neurophysiol.*, vol. 113, no. 6, pp. 767–791, 2002.
- [5] P. Sarma, P. Tripathi, M. P. Sarma, and K. K. Sarma, "Pre-processing and feature extraction techniques for EEG BCI applications—A review of recent research," *ADBU J. Eng. Technol.*, vol. 5, no. 1, 2016.
- [6] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve BCI designs: Unified theory and new algorithms," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 2, pp. 355–362, Feb. 2010.

- [7] J. Müller-Gerking, G. Pfurtscheller, and H. Flyvbjerg, "Designing optimal spatial filters for single-trial EEG classification in a movement task," *Clin. Neurophysiol.*, vol. 110, no. 5, pp. 787–798, 1999.
- [8] S. Lemm, B. Blankertz, G. Curio, and K.-R. Müller, "Spatio-spectral filters for improving the classification of single trial EEG," *IEEE Trans. Biomed. Eng.*, vol. 52, no. 9, pp. 1541–1548, Sep. 2005.
- [9] G. Dornhege, B. Blankertz, M. Krauledat, F. Losch, G. Curio, and K.-R. Müller, "Optimizing spatio-temporal filters for improving brain-computer interfacing," in *Proc. Adv. Neural Inf. Process. Syst.*, 2006, pp. 315–322.
- [10] Q. Novi, C. Guan, T. H. Dat, and P. Xue, "Sub-band common spatial pattern (SBCSP) for brain-computer interface," in *Proc. 3rd Int. IEEE/EMBS Conf. Neural Eng.*, 2007, pp. 204–207.
- [11] K. K. Ang, Z. Y. Chin, H. Zhang, and C. Guan, "Filter bank common spatial pattern (FBCSP) in brain-computer interface," in *Proc. IEEE Int. Joint Conf. Neural Netw. World Congr. Comput. Intell.*, 2008, pp. 2390–2397.
- [12] V. Khare, J. Santhosh, S. Anand, and M. Bhatia, "Brain computer interface based real time control of wheelchair using electroencephalogram," *Int. J. Soft Comput. Eng.*, vol. 1, no. 5, pp. 2231–2307, 2011.
- [13] K. L. Du and M. Swamy, "Radial basis function networks," *Neural Networks in a Softcomputing Framework*. Cham, Switzerland: Springer, 2006, pp. 251–294.
- [14] H. Higashi and T. Tanaka, "Simultaneous design of Fir filter banks and spatial patterns for EEG signal classification," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 4, pp. 1100–1110, Apr. 2012.
- [15] Y. Zhang, G. Zhou, J. Jin, X. Wang, and A. Cichocki, "Optimizing spatial patterns with sparse filter bands for motor-imagery based brain-computer interface," *J. Neurosci. Methods*, vol. 255, pp. 85–91, 2015.
- [16] A. Mammone, M. Turchi, and N. Cristianini, "Support vector machines," *Wiley Interdisciplinary Rev.: Comput. Statist.*, vol. 1, no. 3, pp. 283–289, 2009.
- [17] S. Kumar, A. Sharma, and T. Tsunoda, "An improved discriminative filter bank selection approach for motor imagery EEG signal classification using mutual information," *BMC Bioinform.*, vol. 18, no. 16, 2017, Art. no. 545.
- [18] Y. Zhang, Y. Wang, J. Jin, and X. Wang, "Sparse Bayesian learning for obtaining sparsity of EEG frequency bands based feature vectors in motor imagery classification," *Int. J. Neural Syst.*, vol. 27, no. 2, 2017, Art. no. 1650032.
- [19] P. Pudil, J. Novovičová, and J. Kittler, "Floating search methods in feature selection," *Pattern Recognit. Lett.*, vol. 15, no. 11, pp. 1119–1125, 1994.
- [20] J. Pyllkkönen, "LDA based feature estimation methods for LVCSR," in *Proc. 9th Int. Conf. Spoken Lang. Process.*, 2006, pp. 389–392.
- [21] B. Blankertz, "Data Set IVa for the BCI Competition III." April 15, 2019. Distributed by Intelligent Data Analysis Group, Fraunhofer FIRST, Neurophysics Group, Charit University Medicine Berlin. [Online]. http://www.bbci.de/competition/iii/desc_IVa.html



Muhammad Hamza Bhatti received the bachelor's degree in electrical engineering from the FAST National University of Computer and Emerging Sciences (FAST-NUCES), Lahore, Pakistan, in 2017. He is currently working toward the master's degree in telecommunication and computer networks with the University of Engineering and Technology (UET), Lahore.

Currently he is also working as a Research Officer with the Al-Khawarizmi Institute of Computer Science (KICS), UET. His current research interests include brain computer interface, signal and image processing, machine learning, artificial intelligence, and Internet of Things.



Javeria Khan received the bachelor's degree in mechatronics and control engineering and the master's degree in mechanical design engineering from the University of Engineering and Technology (UET) Lahore, Pakistan, in 2015 and 2018, respectively.

She is currently working as a Research Officer with the Al-Khawarizmi Institute of Computer Science (KICS), UET.



Muhammad Usman Ghani Khan received the Ph.D. degree in statistical modeling for machine vision signals, specifically language descriptions of video streams from Sheffield University, Sheffield, U.K., in 2012.

He is currently an Associate Professor with the Department of Computer Science, University of Engineering and Technology, Lahore, Pakistan. He research interests include spoken language processing using statistical approaches, with applications, such as information extraction from speech and speech summarization. His recent work is concerned with multimedia, incorporating text, audio and visual processing into one frame work.



Razi Iqbal received the master's and Ph.D. degrees in computer science and engineering from Akita University, Akita, Japan, in 2008 and 2011, respectively.

He is an Associate Professor with the College of Computer Information Technology, American University in the Emirates (AUE), Dubai, United Arab Emirates. Prior to joining AUE, he served as the Chairman of the Department of Computer Science and IT, Director of the Office of Research, Innovation, and Commercialization, and as a Research Scientist.



Moayad Aloqaily received the M.Sc. degree in electrical and computer engineering from Concordia University, Montreal, QC, Canada, in 2012, and the Ph.D. degree in electrical and computer engineering from the University of Ottawa, Ottawa, ON, Canada, in 2016.

He is an Instructor with the Systems and Computer Engineering Department at Carleton University, Ottawa. He is also working with Gnowit Inc. Ottawa, ON, as a Senior Researcher and Data Scientist, since 2016. His current research interests include artificial intelligence and machine learning, connected vehicles, blockchain solutions, and data management.

Dr. Aloqaily is a Professional Engineer of Ontario.



Yaser Jararweh received the Ph.D. in computer engineering from the University of Arizona, Tucson, AZ, USA, in 2010.

He is currently an Associate Professor of Computer Science with the Jordan University of Science and Technology, Irbid, Jordan. He has authored and coauthored several technical papers in established journals and conferences in fields related to cloud computing, edge computing, software-defined networking, and big data.



Brij Gupta received the Ph.D. degree in information and cyber security from the Indian Institute of Technology Roorkee, Roorkee, India, in 2011.

He has authored and coauthored more than 70 research papers (including one book and eight chapters).

Dr. Gupta was the recipient of the Canadian Commonwealth Scholarship and awarded the Government of Canada Award, in 2009.